

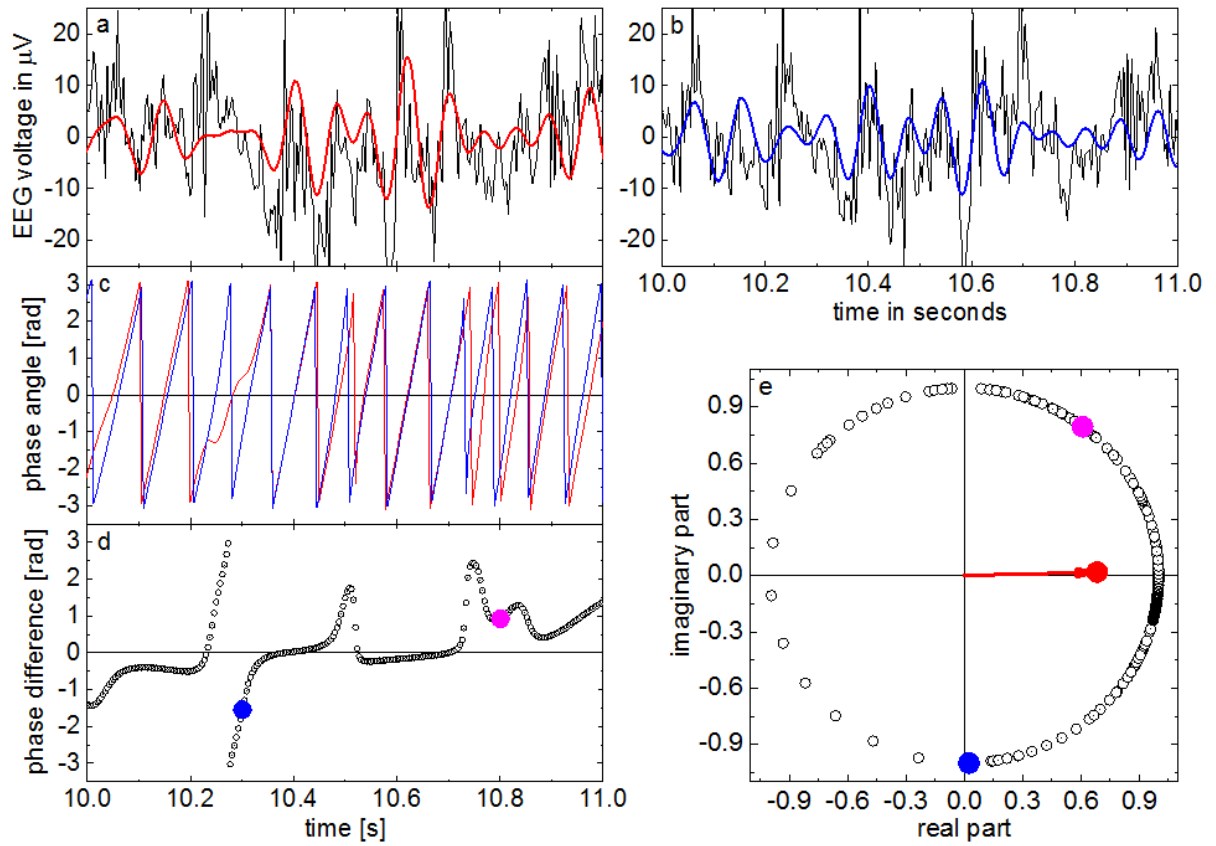


Figure 5: Illustration of EEG phase synchronization. (a,b) By band-pass filtering of the pre-processed EEG signals from two electrodes (artifacts removed, black in the top panels), the oscillations in the alpha band (7.8-15.6 Hz) are retained (red and blue in the top panels). One second sections of these signals for the C3 lead (a) and C4 lead (b) are shown for a PD patient during still standing. Next, the instantaneous phases of both signals are calculated using a Hilbert transform and applying an $\arctan2$ function to the original and the Hilbert transformed components. (c) The centre left panel shows both phases ϕ_1 and ϕ_2 (red and blue as in a and b; range $-\pi = -180^\circ$ to $+\pi = +180^\circ$). (d) The difference between both phases $\Delta\phi = \phi_2 - \phi_1$ is shown in the bottom left panel for the same time interval. Two points are marked blue and magenta for comparison with panel (e). (e) Finally, complex exponentials of the phase difference, $\exp[i \Delta\phi]$ (i.e. numbers on a unit circle in the complex plane), are averaged over intervals of 10 oscillations to obtain the red dot as average value. In this case, more points appear on the right-hand side of the circle, and therefore the red dot is towards the right. The distance of this dot from the plot origin (i.e., the length of the red arrow) is the phase synchronization index ranging from 0 to 1; in this example it is 0.38, indicating weak phase synchronization.